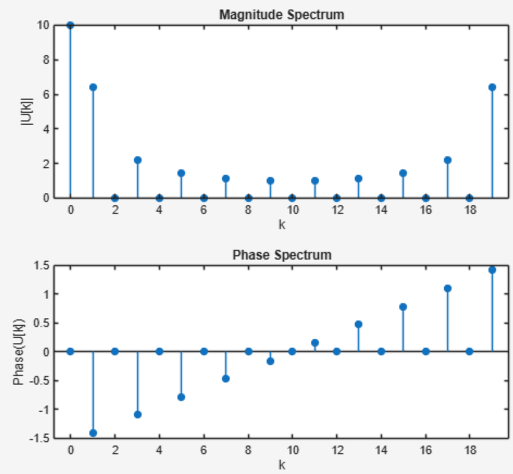
lab6task1.m

/MATLAB Drive/lab6task1.m

```
clc;clear;close all;
N=10;
M=20;
n=0:N-1;
u=ones(1,N);
U=fft(u,M);
k=0:M-1;
figure;
subplot(2,1,1);
stem(k,abs(U),'filled');
title('Magnitude Spectrum');
xlabel('k');
ylabel('|U[k]|');
subplot(2,1,2);
stem(k,angle(U),'filled');
title('Phase Spectrum');
xlabel('k');
ylabel('Phase(U[k])');
```

Figure 1



lab6task1.m lab6task2.m

/MATLAB Drive/lab6task2.m

```
clc;clear;close all;
x1=[1,2,3,4];
x2=[4,3,2,1];
N=length(x1);
a=2;b=3;
X1=fft(x1);
X2=fft(x2);
LHS_Lin=fft(a*x1+b*x2);
RHS_Lin=a*X1+b*X2;
disp(['Linearity Error: ',num2str(norm(LHS_Lin-RHS_Lin))]);
LHS_TS=fft(circshift(x1,1));
RHS_TS=X1.*exp(-1j*2*pi*(0:N-1)*1/N);
disp(['Time Shift Error: ',num2str(norm(LHS_TS-RHS_TS))]);
LHS_FS=fft(x1.*exp(1j*2*pi*(0:N-1)*1/N));
RHS_FS=circshift(X1,1);
disp(['Freq Shift Error: ',num2str(norm(LHS_FS-RHS_FS))]);
LHS_Conv=fft(cconv(x1,x2,N));
RHS_Conv=X1.*X2;
disp(['Circ Convolution Error: ',num2str(norm(LHS_Conv-RHS_Conv))]);
LHS_Mult=fft(x1.*x2);
RHS_Mult=cconv(X1,X2,N)/N;
disp(['Multiplication Error: ',num2str(norm(LHS_Mult-RHS_Mult))]);
E_time=sum(abs(x1).^2);
E_freq=sum(abs(X1).^2)/N;
disp(['Parseval Error: ',num2str(abs(E_time-E_freq))]);
```

Command Window

```
Linearity Error: 0
Time Shift Error: 7.0974e-16
Freq Shift Error: 1.6659e-15
Circ Convolution Error: 0
Multiplication Error: 0
Parseval Error: 0
```

lab6task1.m × lab6task2.m × lab6task3.m × +

/MATLAB Drive/lab6task3.m

```
clc;clear;close all;
g=[3,4,-2,0,1,-4];
h=[1,-3,0,4,-2,3];
N_a=length(g);
circ_conv_a=ifft(fft(g).*fft(h));
disp('Circular convolution result for (a):');
disp(circ_conv_a);
n=0:4;
RollNumber=10;
x=sin(pi*n/2);
y=RollNumber.^n;
circ_conv_b=ifft(fft(x).*fft(y));
disp('Circular convolution result for (b):');
disp(circ_conv_b);
```

Command Window

```
Circular convolution result for (a):
   31.0000   -7.0000  -32.0000   29.0000   -1.0000  -14.0000

Circular convolution result for (b):
   1.0e+03 *

    9.9000   -0.9990   -9.9900    0.0990    0.9900
```